TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – DBMS

Lecture No.- 7

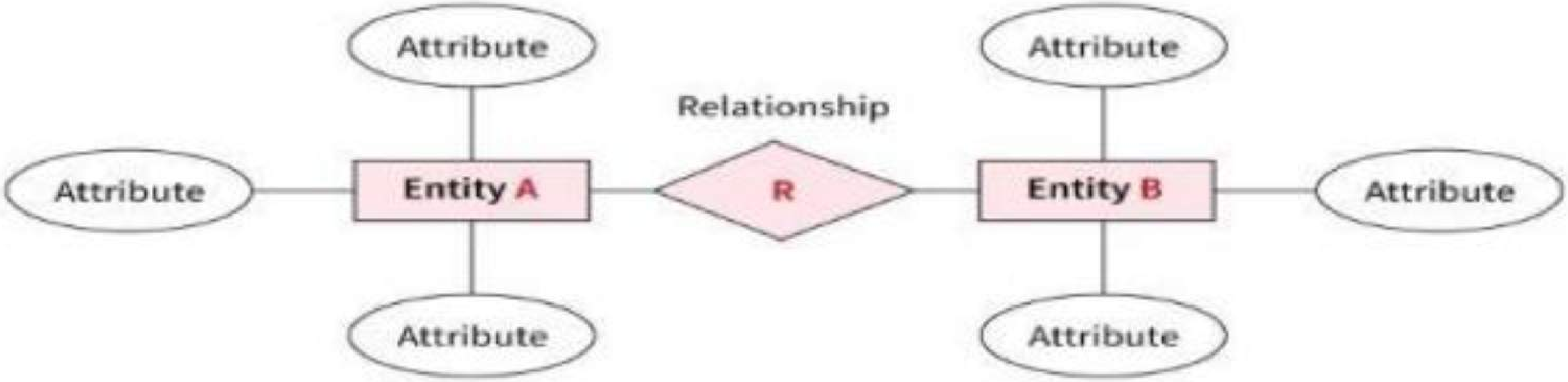
By- Rahul Sir

Topics to be Covered

1:- ER-MODEL
2:- Various Types of Entity
3:- Various Types of Attribute
4:- Various Types of Relations
5:- Discussion on MCQ problem
6:- Basic Discussion on Normalization

Topic :ER model

ER model in DBMS is the high-level data model. It stands for the **Entity-relationship model** and is used to **represent a logical view of the system(Conceptual view)** from a data perspective.
An ER diagram in DBMS defines entities, associated attributes,
and relationships between entities. This helps visualize the logical structure of the database. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.



Why Use ER Diagrams In DBMS ?

1:-ER diagrams represent the E-R model in a database, making them easy to convert into relations (tables).

2:-ER diagrams serve the purpose of real-world modeling of objects which makes them intently useful.

3:-ER diagrams require no technical knowledge of the underlying DBMS used.

4:-It gives a standard solution for visualizing the data logically.

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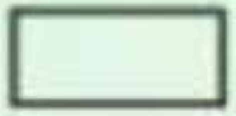
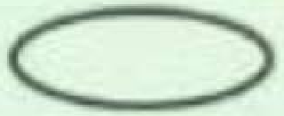
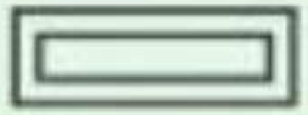
Symbols Used in ER Model

ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:

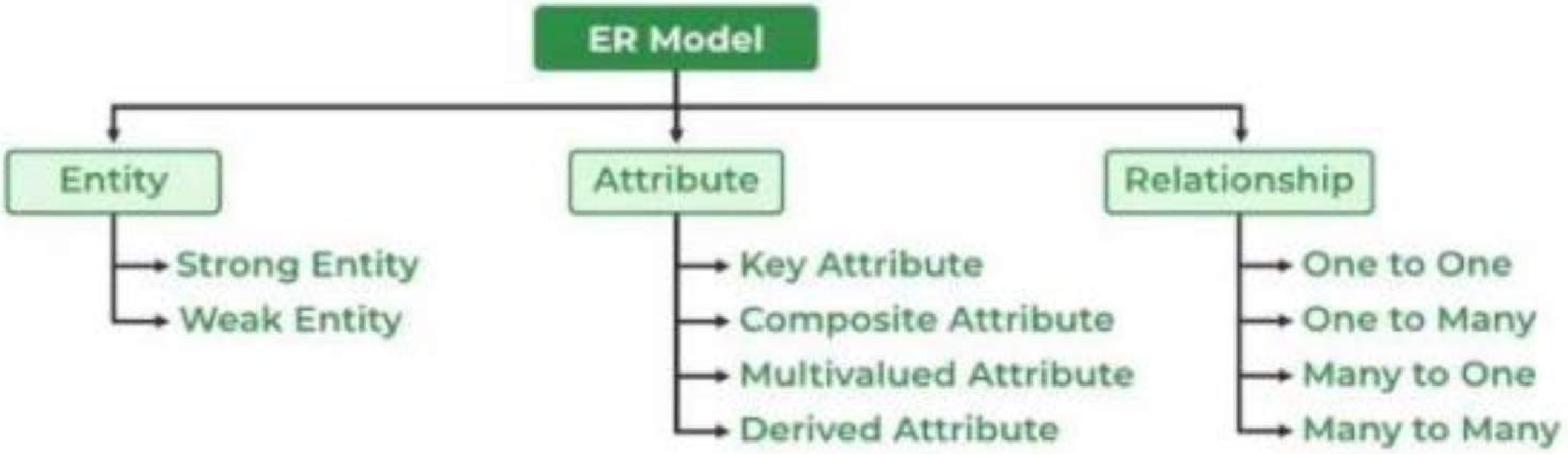
- **Rectangles:** Rectangles represent entities in the ER Model.
- **Ellipses:** Ellipses represent attributes in the ER Model.
- **Diamond:** Diamonds represent relationships among Entities.
- **Lines:** Lines represent attributes to entities and entity sets with other relationship types.
- **Double Ellipse:** Double ellipses represent multi-valued Attributes.
- **Double Rectangle:** Double rectangle represents a weak entity.

Symbols Used in ER Model

Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Components of ER Diagram

ER Model consists of Entities, Attributes, and Relationships among Entities in a Database System.

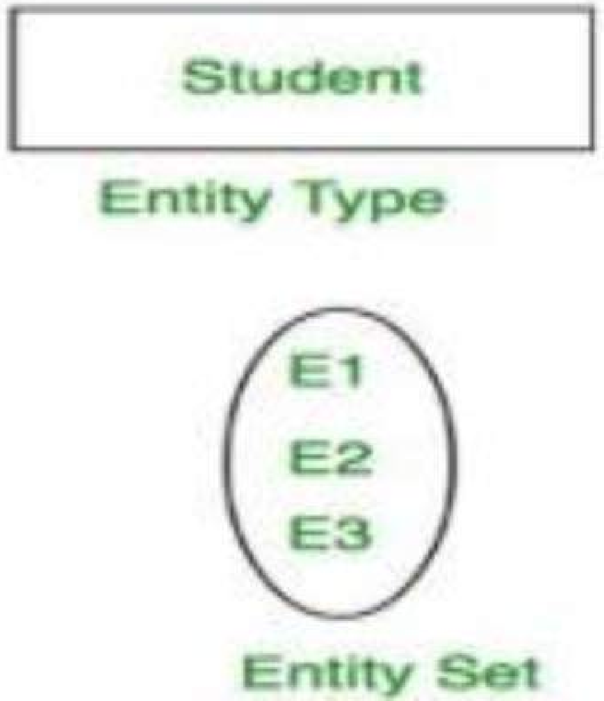


What is an Entity

An Entity may be an object with a physical existence: a particular person, car, house, or employee or it may be an object with a conceptual existence – a company, a job, or a university course

What is an Entity Set

An entity refers to an individual object of an entity type, and the collection of all entities of a particular type is called an entity set. For example, E1 is an entity that belongs to the entity type "Student," and the group of all students forms the entity set. In the ER diagram below, the entity type is represented as:



Types of Entity

There are two types of entity:

1. Strong Entity
A Strong Entity is a type of entity that has a key Attribute. Strong Entity does not depend on other Entity in the Schema. It has a primary key, that helps in identifying it uniquely, and it is represented by a rectangle. These are called Strong Entity Types.

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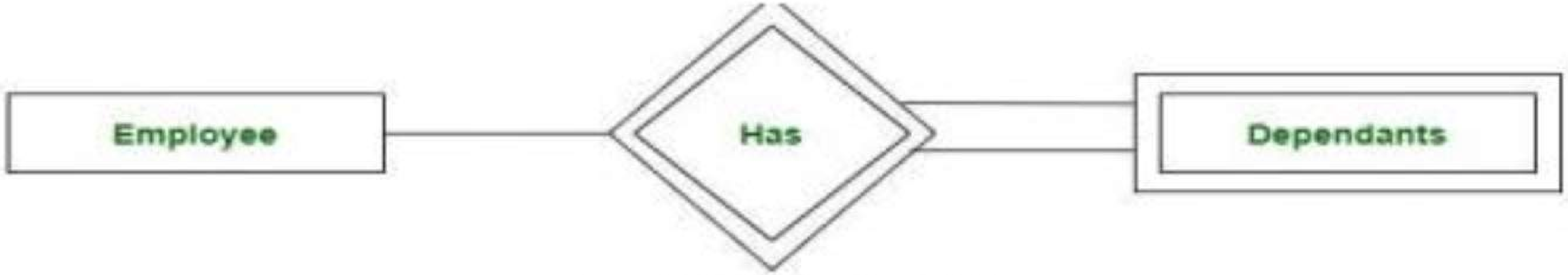
Types of Entity

2. Weak Entity

An Entity type has a key attribute that uniquely identifies each entity in the entity set. But some entity type exists for which key attributes can't be defined. These are called Weak Entity types.

For Example, A company may store the information of dependents (Parents, Children, Spouse) of an Employee. But the dependents can't exist without the employee. So dependent will be a **Weak Entity Type** and Employee will be identifying entity type for dependent, which means it is **Strong Entity Type.**

A weak entity type is represented by a double rectangle. The participation of weak entity types is always total. The relationship between the weak entity type and its identifying strong entity type is called identifying relationship and it is represented by a double diamond.



Strong Entity and Weak Entity

Topic : ER model

What are Attributes

Attributes are the properties that define the entity type. For example, Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.

Types of Attributes

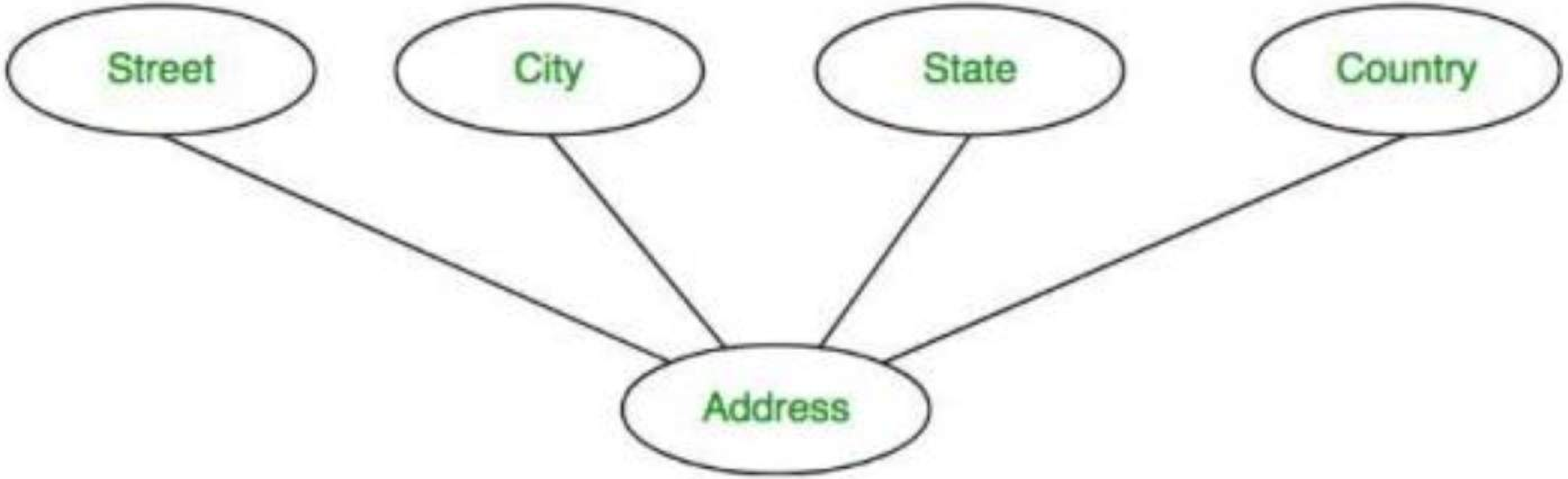
1. Key Attribute

The attribute which **uniquely identifies each entity** in the entity set is called the key attribute. For example, Roll_No will be unique for each student. In ER diagram, the key attribute is represented by an oval with underlying lines.

Key Attribute

2. Composite Attribute

An attribute **composed of many other attributes** is called a composite attribute. For example, the Address attribute of the student Entity type consists of Street, City, State, and Country. In ER diagram, the composite attribute is represented by an oval comprising of ovals.



Composite Attribute

3. Multivalued Attribute

An attribute consisting of more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, a multivalued attribute is represented by a double oval.

Multivalued Attribute

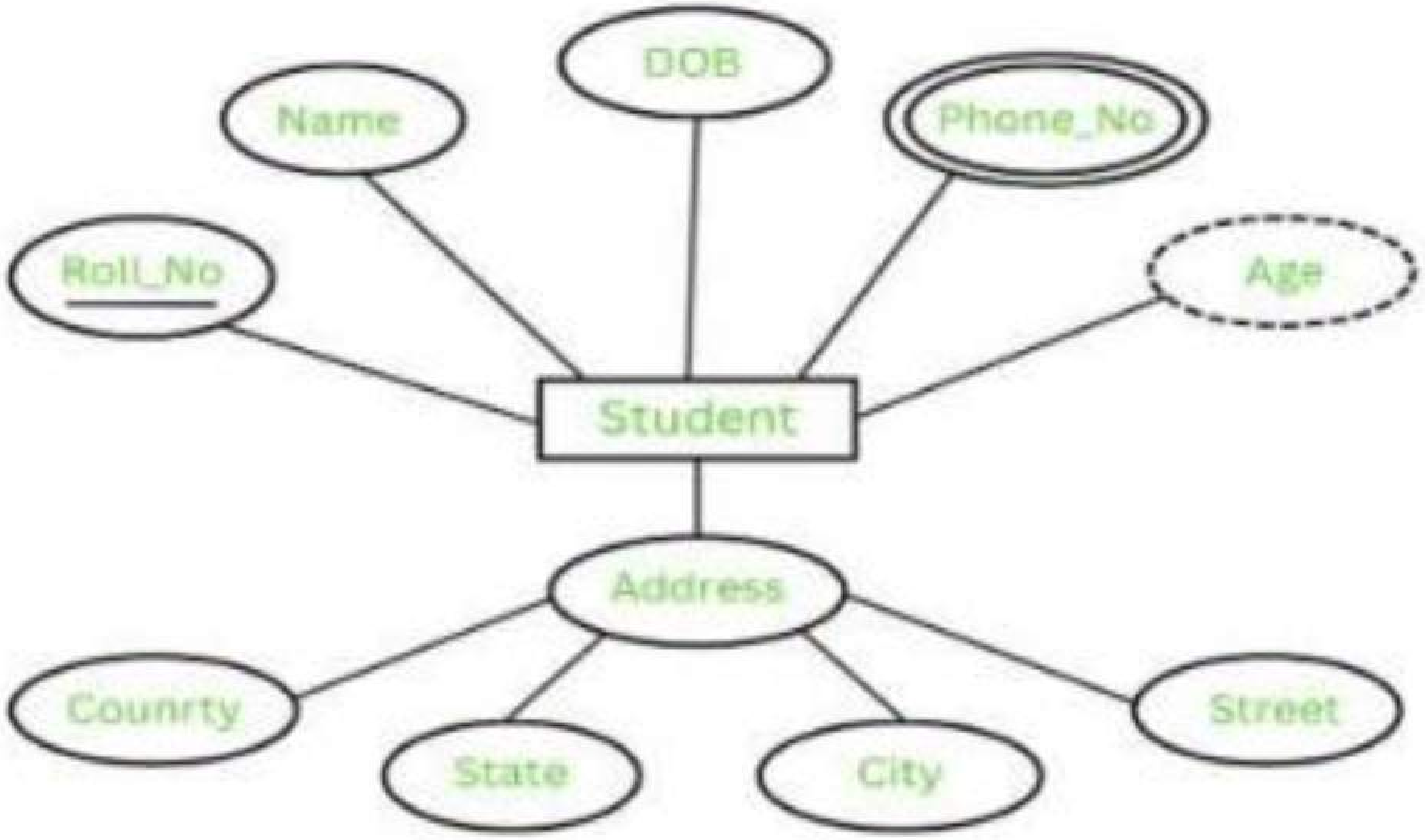
4. Derived Attribute

An attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, the derived attribute is represented by a dashed oval.

Derived Attribute

Topic : ER model

The Complete Entity Type Student with its Attributes can be represented as:

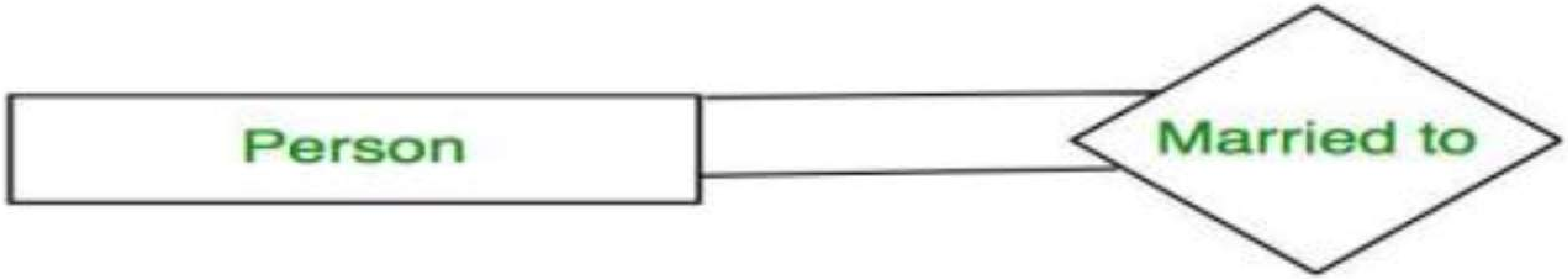


Entity and Attributes

Degree of a Relationship Set

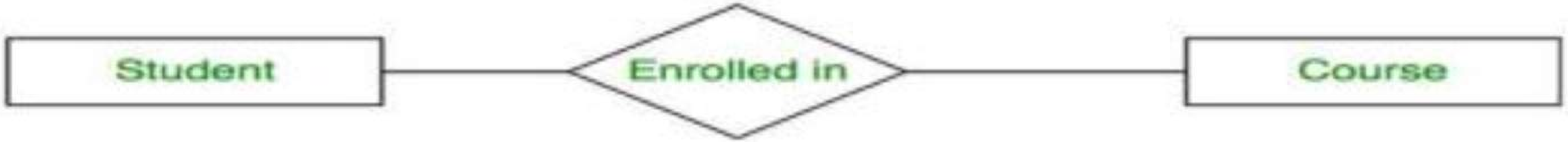
The number of different entity sets participating in a relationship set is called the degree of a relationship set.

1. Unary Relationship: When there is only ONE entity set participating in a relation, the relationship is called a unary relationship. For example, one person is married to only one person.



Unary Relationship

2. Binary Relationship: When there are TWO entities set participating in a relationship, the relationship is called a binary relationship. For example, a Student is enrolled in a Course.



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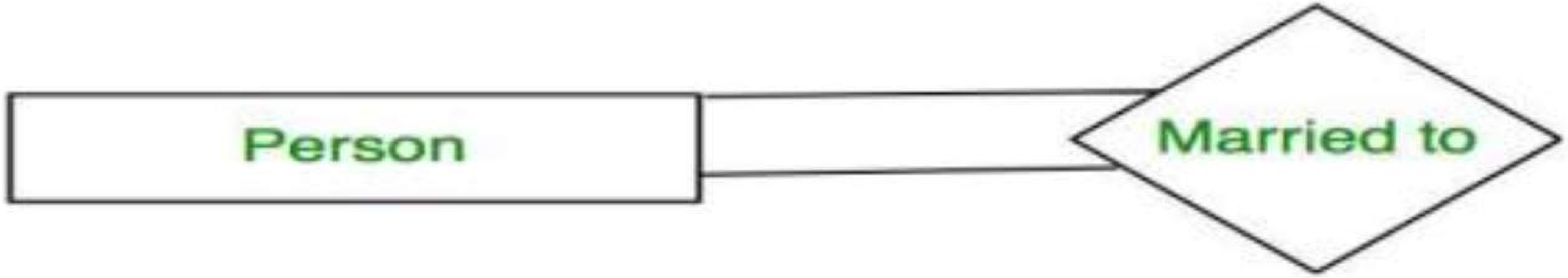

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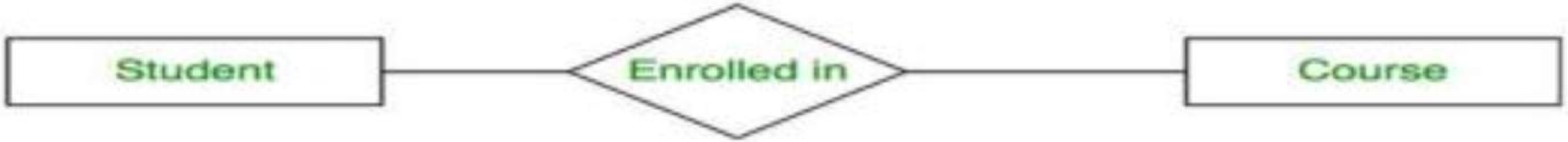


Binary Relationship

3. Ternary Relationship: When there are three entity sets participating in a relationship, the relationship is called a ternary relationship.

4. N-ary Relationship: When there are n entities set participating in a relationship, the relationship is called an n-ary relationship.

What is Cardinality

The maximum number of times an entity of an entity set participates in a relationship set is known as cardinality . Cardinality can be of different types:

1:- One-to-One: When each entity in each entity set can take part only once in the relations.

Topic : ER model

FK 1 FK 2

EID	ENAME	AGE
E1	RAM	20
E2	SHYAM	25
E3	RAM	28
E4	RAVI	24

Parent (Referenced table)

EID	DID
E1	D1
E3	D2
E4	D3
E2	D4

Child (Referencing table)

DID	DNAME	LOCATION
D1	IT	BANGLORE
D2	PRODUCTION	DELHI
D3	HR	MIRZAPUR
D4	MARKETING	LUCKNOW

Parent 2 (Referenced table)

Here you can Create any one Column primary key Either

EID OR DID

Primary key 1 foreign key 1 → primary key 2

EID	ENAME	AGE	DID
E1	RAM	20	D1
E2	SHYAM	25	D4
E3	RAM	28	D2
E4	RAVI	24	D3

DID	DNAME	LOCATION
D1	IT	BANGLORE
D2	PRODUCTION	DELHI
D3	HR	MIRZAPUR
D4	MARKETING	LUCKNOW

It is Reduce in to Two table

2. **One-to-Many:** In one-to-many mapping as well where each entity can be related to more than one entity.

Parent 1 (Referenced table)

CID	ENAME	CITY
C1	RAM	DELHI
C2	SHYAM	MUMBAI
C3	RAM	MIRZAPUR
C4	RAVI	VARANSI

FK 1 FK 2

Child (Referencing table)

CID	OID
C1	O1
C1	O2
C2	O3
C2	O4

Parent 2 (Referenced table)

OID	DNAME	COST
O1	SHIRT	1000
O2	T-SHIRT	500
O3	KURTA	1500
O4	PANT	800

Here you can Create Only one Column primary key

OID

2. **One-to-Many:** In one-to-many mapping as well where each entity can be related to more than one entity.

Primary key 1

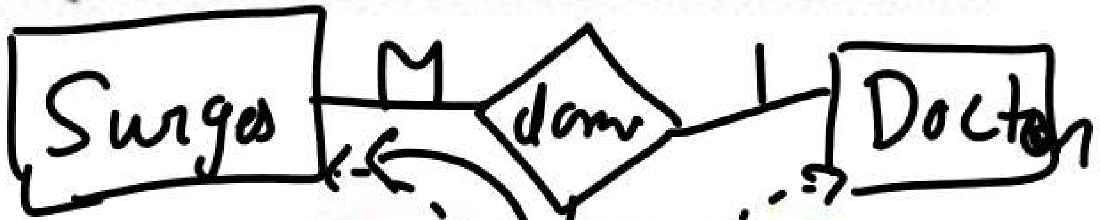
CID	ENAME	CITY
C1	RAM	DELHI
C2	SHYAM	MUMBAI
C3	RAM	MIRZAPUR
C4	RAVI	VARANSI

FK 1 PK 2

CID	OID	DNAME	COST
C1	O1	SHIRT	1000
C1	O2	T-SHIRT	500
C2	O3	KURTA	1500
C2	O4	PANT	800

It is Reduce in to Two table

3. Many-to-One: When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in the relationship set, cardinality is many to one..



OID OID2

Parent 2 (Referenced table)

CID	ENAME	CITY
C1	RAM	DELHI
C2	SHYAM	MUMBAI
C3	RAM	MIRZAPUR
C4	RAVI	VARANSI

Parent 1 (Referenced table)

OID1	OID2
O1	O2
O3	O4
null	null
null	null

FK 1 FK 2

CID	OID
C1	O1
C1	O2
C2	O3
C2	O4

Child (Referencing table)

OID	DNAME	COST
O1	SHIRT	1000
O2	T-SHIRT	500
O3	KURTA	1500
O4	PANT	800

Here you can Create Only one Column primary key

OID

3. Many-to-One: When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in the relationship set, cardinality is many to one..

Primary key 1

CID	ENAME	CITY
C1	RAM	DELHI
C2	SHYAM	MUMBAI
C3	RAM	MIRZAPUR
C4	RAVI	VARANSI

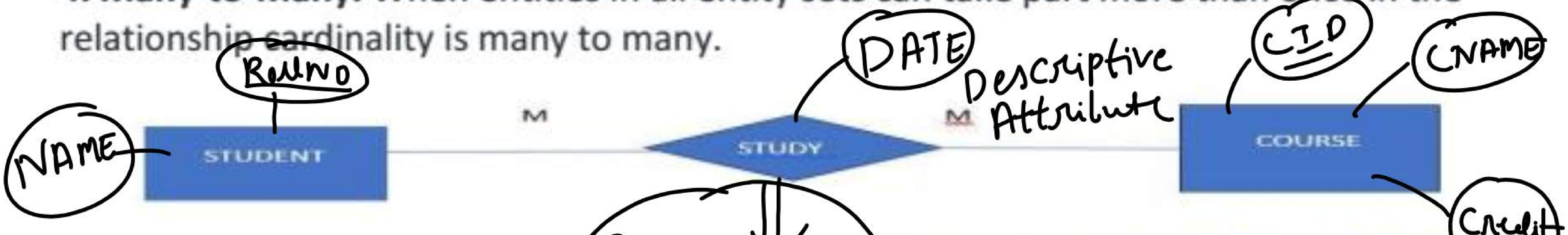
FK 1 PK 2

CID	OID	DNAME	COST
C1	O1	SHIRT	1000
C1	O2	T-SHIRT	500
C2	O3	KURTA	1500
C2	O4	PANT	800

It is Reduce in to Two table

Topic : ER model

4. Many-to-Many: When entities in all entity sets can take part more than once in the relationship cardinality is many to many.



Composite key (RollNo+CID)

ROLLNO	NAME	AGE	CID
1	RAM	16	C1, C2
2	SHYAM	17	
3	RAM	16	
4	RAVI	17	

Parent 1 (Referenced table)

FK 1	FK 2	
ROLL NO	CID	DATE
1	C1	
2	C2	
1	C2	
3	C3	

Child (Referencing table)

Parent 2 (Referenced table)

CID	CNAME	CREDIT	Roll
C1	MATHS	4	1
C2	PHYSICS	4	1,2
C3	CHEMISTRY	4	
C4	COMPUTER	3	

Here We can not Create primary key but We can Create Composite key (ROLL NO + CID)

Note:- Here We can not reduce 3 table in to 2

Topic : MCQ ON ER-MODEL

1. An ________ is a set of entities of the same type that share the same properties, or attributes.
   a) Entity set
   b) Attribute set
   c) Relation set
   d) Entity model

2. Entity is a ________
   a) Object of relation
   b) Present working model
   c) Thing in real world
   d) Model of relation

3. The descriptive property possessed by each entity set is ________
   a) Entity
   b) Attribute
   c) Relation
   d) Model

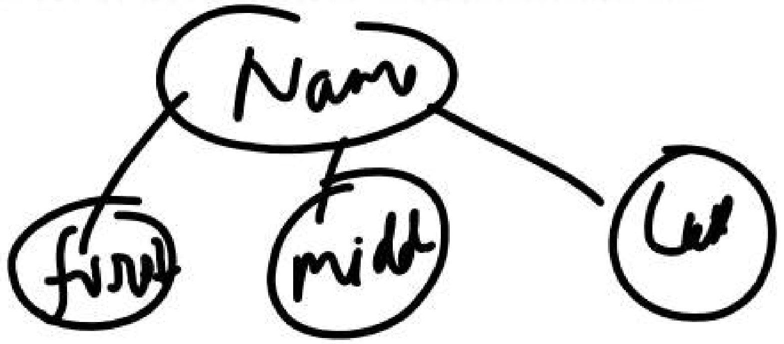
Topic : MCQ ON ER-MODEL

4. The function that an entity plays in a relationship is called that entity's ____________

   a) Participation
   b) Position
   c) Role
   d) Instance

5. The attribute *name* could be structured as an attribute consisting of first name, middle initial, and last name. This type of attribute is called

   a) Simple attribute
   b) Composite attribute
   c) Multivalued attribute
   d) Derived attribute



6. The attribute AGE is calculated from DATE_OF_BIRTH. The attribute AGE is

   a) Single valued
   b) Multi valued
   c) Composite
   d) Derived

7. Not applicable condition can be represented in relation entry as
   a) NA
   b) 0
   c) NULL
   d) Blank Space

8. Which of the following can be a multivalued attribute?
   a) Phone_number
   b) Name
   c) Date_of_birth
   d) All of the mentioned

9. 9. Which of the following is a single valued attribute
   a) Register_number
   b) Address
   c) SUBJECT_TAKEN
   d) Reference

Topic : MCQ ON ER-MODEL

10. 10. In a relation between the entities the type and condition of the relation should be specified. That is called as ________ attribute.

a) Desciptive
b) Derived
c) Recursive
d) Relative

11.

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions can the relational table for R be merged with that of A?

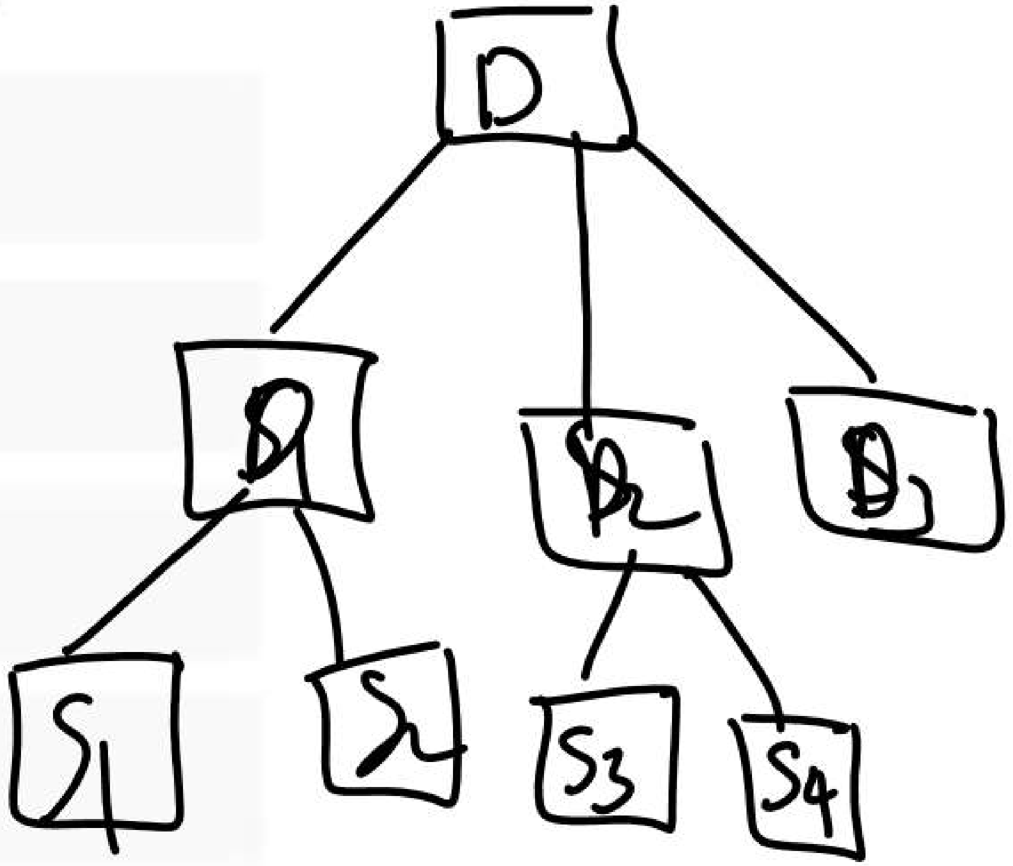
1. Relationship R is one-to-many and the participation of A in R is total
2. Relationship R is one-to-many and the participation of A in R is partial
3. Relationship R is many-to-one and the participation of A in R is total
4. Relationship R is many-to-one and the participation of A in R is partial

Topic : MCQ ON ER-MODEL

12.

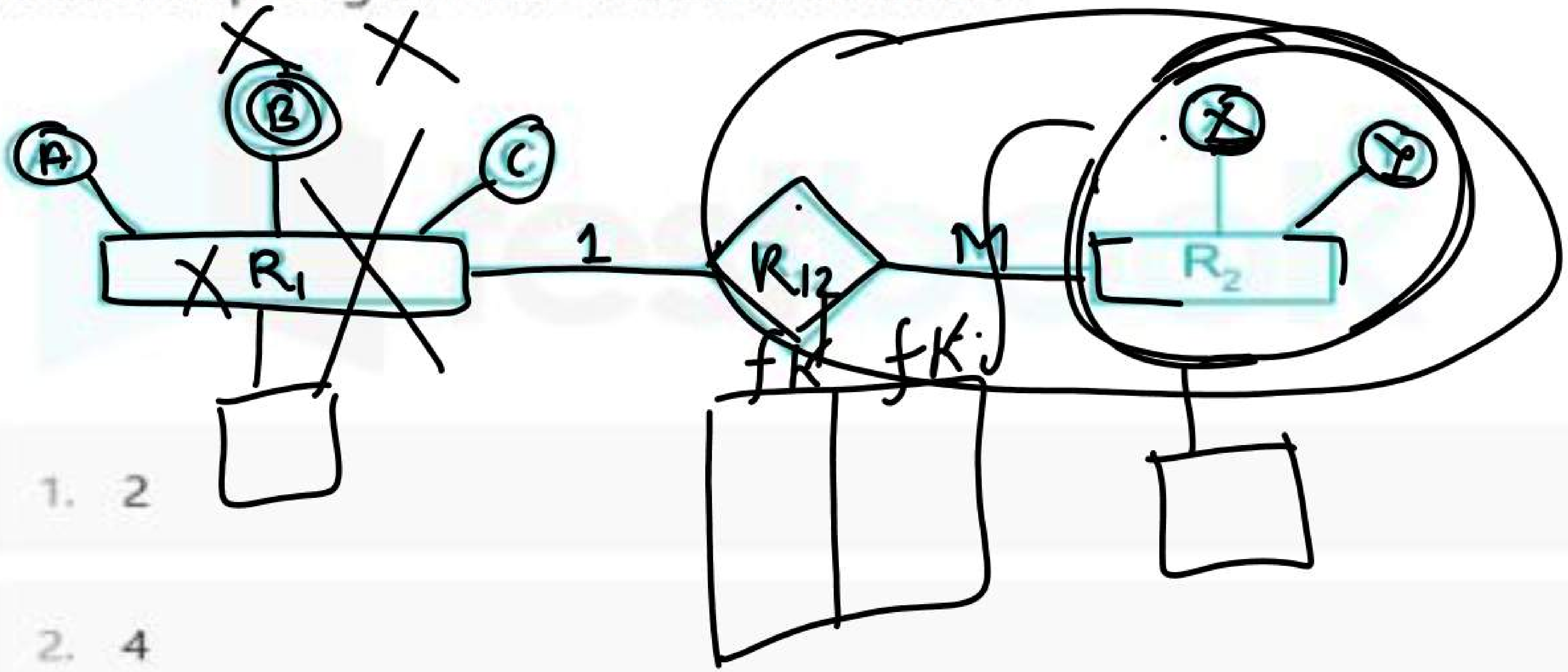
The logical data structure with a one-to-many relationship is a:

1. Network
2. Tree ✓
3. Chain
4. Relationship
5. Schema

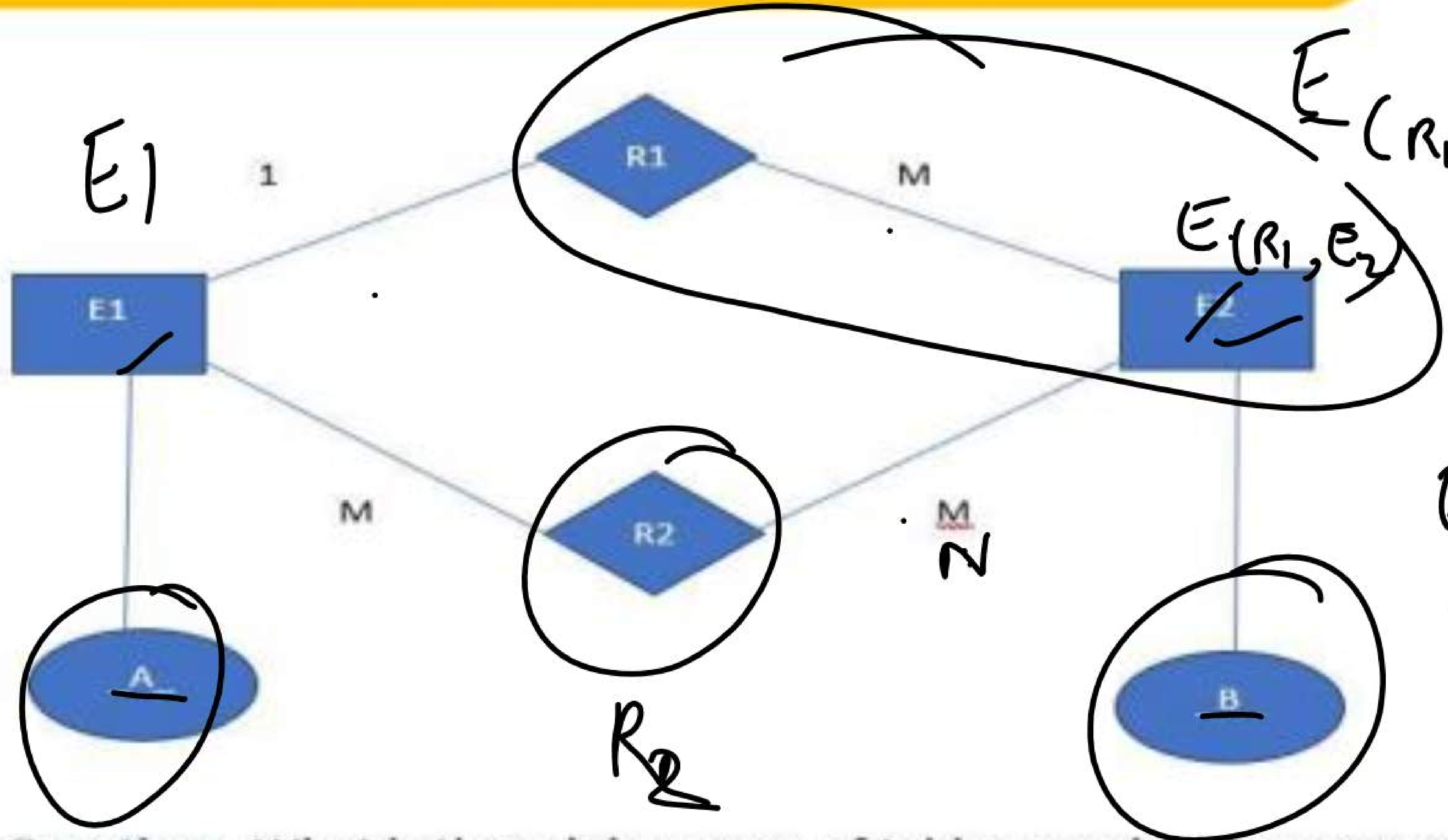


13.

Find minimum number of tables required for converting the following entity relationship diagram into relational database?



1. 2
2. 4
3. 3 ✓



$$\frac{E_1, R_2, E_{(R_1,E_2)}}{3}$$

Question:- What is the minimum no of tables required to represent this E-R MODEL in to Relational Model ?

(a) 2 (b) 3 (c) 4 (d) 5

Normalization is the process of organizing the data and the attributes of a database. It is performed to reduce the data redundancy in a database and to ensure that data is stored logically. Data redundancy in DBMS means having the same data but at multiple places. It is necessary to remove data redundancy because it causes anomalies in a database which makes it very hard for a database administrator to maintain it.

Row Wise Redundancy:-

SID	SNAME	AGE
1	RAVI	20
2	SURYA	23
1	RAVI	21

Topic :Normalization

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COLLEGE MANAGEMENT

Column wise Redundancy:-

STUDENT		COURSE		FACULTY		
SID	SNAME	CID	CNAME	FID	FNAME	SALARY
1	RAM	C1	DBMS	F1	JOHN	30000
2	SURYA	C2	JAVA	F2	RAVI	40000
3	NITIN	C1	DBMS	F1	JOHN	30000
4	AMRIT	C1	DBMS	F1	JOHN	30000

Topic :Normalization

Why Do We Need Normalization?

As we have discussed above, normalization is used to reduce data redundancy. It provides a method to remove the following anomalies from the database and bring it to a more consistent state:

A database anomaly is a flaw in the database that occurs because of poor planning and redundancy.

1:-**Insertion anomalies**: This occurs when we are not able to insert data into a database because some attributes may be missing at the time of insertion.

2:-**Updation anomalies:** This occurs when the same data items are repeated with the same values and are not linked to each other.

3:-**Deletion anomalies:** This occurs when deleting one part of the data deletes the other necessary information from the database.

THANK YOU